

Experiment (8)

CONSERVATION OF MECHANICAL ENERGY (MAXWELL'S WHEEL)

8.1 Introduction

Mechanical energy is the energy an object has because of its motion (Kinetic Energy) or position (Potential Energy). The sum of these energies is conserved for an isolated system if it is exposed only to conservative forces. This conservation principle must be accounted for in the design and execution of any process, whether in research, industry, or agriculture. Maxwell's Wheel, as it falls and rises, will continually convert mechanical energy between various forms. Let us test whether, indeed, the sum of these different forms will be conserved throughout this process.

8.2 Objectives

- Observe interconversion between forms of mechanical energy (work, translational kinetic energy, rotational kinetic energy, gravitational potential energy).
- Explore concepts of rotational motion (angular velocity, moment of inertia).
- Verify the law of conservation of energy for conservative forces using Maxwell's wheel.

8.3 Theory

The law of conservation of mechanical energy can be demonstrated using Maxwell's wheel. It consists of a disc of radius R having an axis of radius r ($r \ll R$) suspended from a fixed frame by two strings of equal length, which are wound around the axis as shown in Figure 8.1.



Figure 8.1 : Maxwell's wheel

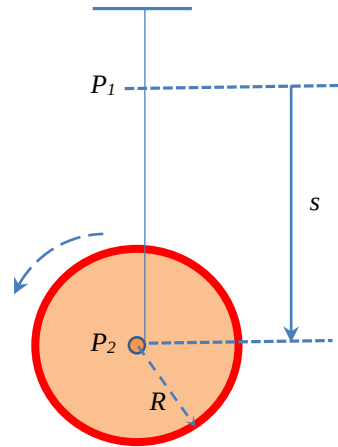


Figure 8.2 : Maxwell's wheel diagram

If we release the wheel to rotate and fall a distance s from point p_1 to point p_2 , as shown in Figure 8.2 above, the gravitational force (which can be considered to act on the center of mass of the wheel) does a positive work W at the expense of the gravitational potential energy E_p (which decreases by the same amount). This work is transformed into translational kinetic energy E_T and rotational kinetic energy E_r .i.e.,

$$m g s = E_T + E_r = \frac{1}{2} m v^2 + \frac{1}{2} I \omega^2 \quad (8.1)$$

Where g is the gravitational acceleration, m is the mass of the wheel and its axis, I is the moment of inertia of the wheel about its axis and ω is the rotational velocity of the wheel.

Using Figure 8.3 which indicates the string and the rotating axis of the wheel during the fall, the relations between the distance s , the translational velocity v and the rotational velocity ω are given by,

$$v = \frac{ds}{dt} \quad (8.2)$$

$$\omega = \frac{d\theta}{dt} = \frac{1}{r} \frac{rd\theta}{dt} = \frac{1}{r} \frac{ds}{dt} = \frac{v}{r} \quad (8.3)$$

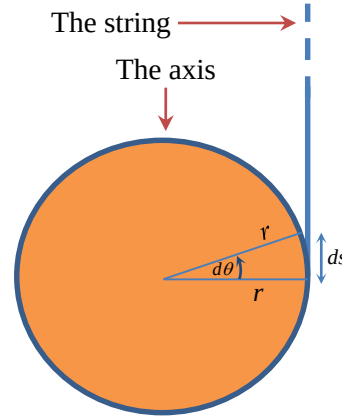


Figure 8.3 : The string and the rotating axis.

In Equation (8.3), we take the limit for an infinitesimally small displacement, ds , for which we can say that $ds = r d\theta$. Then Equation (8.1) can be rewritten as,

$$m g s = \frac{1}{2} \left(m + \frac{I}{r^2} \right) v^2 \quad (8.4)$$

Differentiating this relation with respect to time, we get,

$$m g \frac{ds}{dt} = \frac{1}{2} \left(m + \frac{I}{r^2} \right) 2v \frac{dv}{dt}$$

Noting that

$$v = \frac{ds}{dt}, \text{ and } a = \frac{dv}{dt} \text{ we get,}$$

$$a = \frac{mg}{\left(m + \frac{I}{r^2} \right)}$$

Considering the initial conditions at point p_1 ($t=0$, $s_o = 0$, $v_o = 0$) and the downward direction to be positive , we get,

$$s = \frac{1}{2}at^2 = \frac{1}{2} \frac{mg}{(m + \frac{I}{r^2})} t^2 \quad (8.5)$$

and,

$$v = \frac{ds}{dt} = \frac{mg}{(m + \frac{I}{r^2})} t \quad (8.6)$$

By measuring the velocity v of the wheel at distance s corresponding to time t after releasing the wheel, we can measure the moment of inertia of the wheel about its axis using any of the relations (8.4) , (8.5) or (8.6) .

In addition, we can verify the equation (8.1) of conservation of mechanical energy .

8.4 Setup

Maxwell's wheel experiment main parts are shown in Figure 8.4.

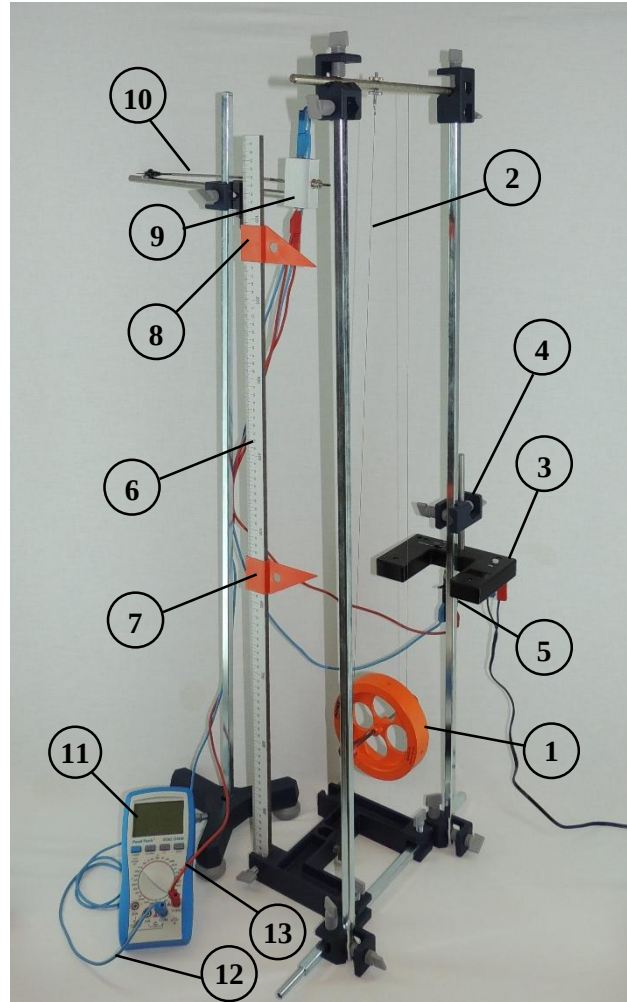


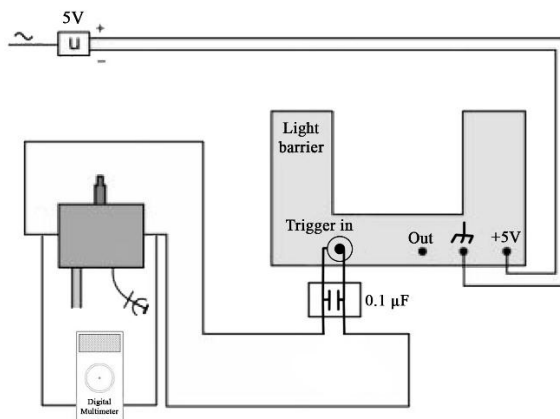
Figure 8.4 : Experimental Setup of Maxwell's wheel.

- | | |
|--|--------------------------|
| 1 Maxwell's wheel | 8 Upper ruler pointer |
| 2 String | 9 Holding device |
| 3 Light barrier with counter | 10 Release trigger |
| 4 Light barrier height adjusting screw | 11 Digital Multimeter |
| 5 Capacitor, $0.1\mu\text{F}$ | 12 Connecting cord, blue |
| 6 Ruler | 13 Connecting cord, red |
| 7 Lower ruler pointer | |

8.5 Procedure

8.5.1 Task (1): Determination of moment of inertia I of the wheel about its axis using $(S - t^2)$ curve (relation (8.5)) and $(v-t)$ curve (relation (8.6)).

- Figure 8.5 (a) and (b) show the electric wiring of the holding device (9) with both the light barrier (3), the capacitor (5) and the digital multimeter (11).



(a)



(b)

Figure 8.5 : Wire connections between the holding device and both the light barrier and the digital multimeter.



Make sure when the string (2) is totally unwound and Maxwell's wheel (1) is stationary at it's lowest position that the wheel axis is aligned horizontally. If not, ask your TA to adjust it.

- Make sure the light barrier (3) is on and set its switch to the third mode (▲▲) as shown in Figure 8.6.

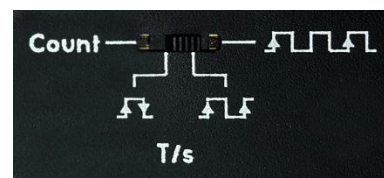


Figure 8.6: Set the light barrier switch to the third mode.

- Hold the wheel's axis with both hands, one at each end, and wind the string (2) on the axis by the rotating the wheel about its axis. The wheel will move

upwards as you wind the string, stop winding when the center of the wheel is at the same horizontal level as the holding device (9).



The windings must run inwards and are wound uniformly on both sides of the axis of Maxwell's wheel as shown in Figure 8.7.

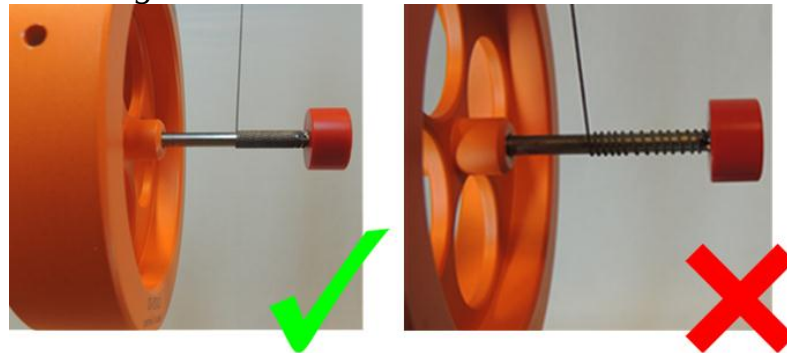


Figure 8.7 : Correct and incorrect winding densities.

- 4- Press the release trigger (10) of the the holding device (9) and lock it as shown in Figure 8.8. You will then be able to use the pin of the holding device and the hole in the circumference of the wheel shown in Figure 8.9 to hold the wheel at the required height and prevent it from descending and the string from unwinding. The holding device should be adjusted such that the wheel does not oscillate or roll after the start.

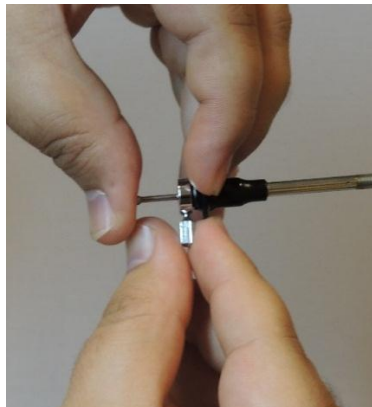


Figure 8.8: Pressing and locking the release trigger.

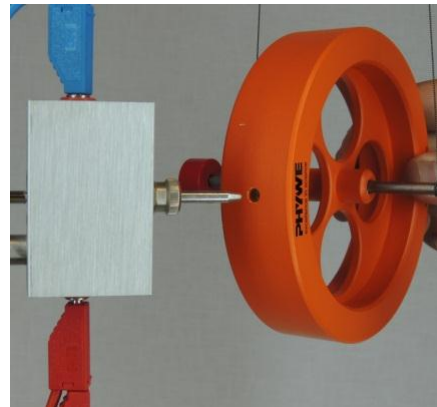


Figure 8.9 : Holding the wheel using the device holder.



The holding device pin should be at the same horizontal level of the center of the wheel. The correct and incorrect ways of holding the wheel using the device holder are shown in Figure 8.10.

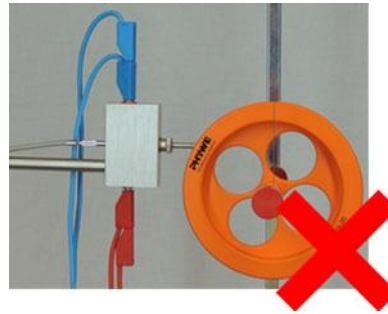
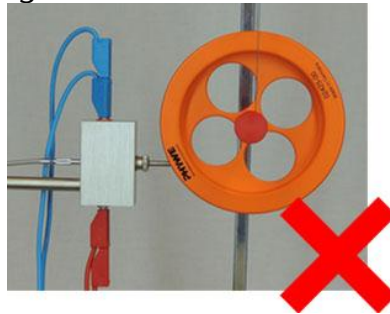
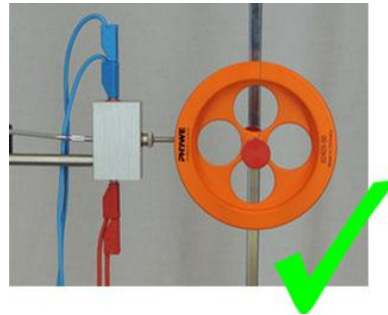


Figure 8.10: Correct and incorrect ways of holding the wheel using the device holder.

- 5- Adjust the distance S covered by Maxwell's wheel measured from the center of the wheel to the middle of the light barrier to be 15 cm by using the ruler (6) and setting the distance between the lower (7) and upper (8) ruler pointers to the required value S . Then use the light barrier height adjusting screw (4) to unscrew and move the light barrier to

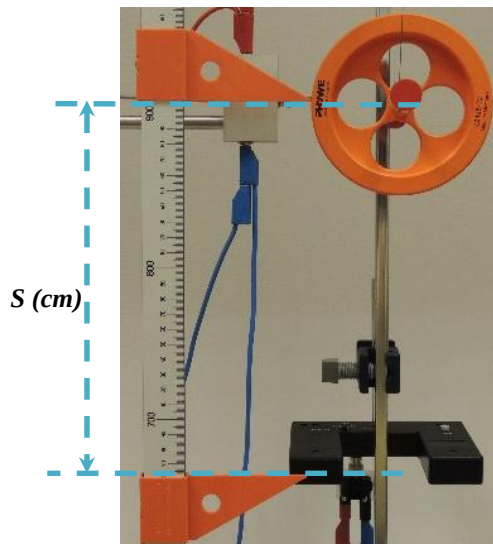


Figure 8.11: Adjusting the distance S .

the same horizontal level as the lower ruler pointer as shown in Figure 8.11.



Use the **screw found at the side** of the light barrier height adjusting screw (4) to be able to adjust the vertical position of the light barrier as shown in Figure 8.12. **DO NOT unscrew the screw found at the back** as it will cause the light barrier to fall.)

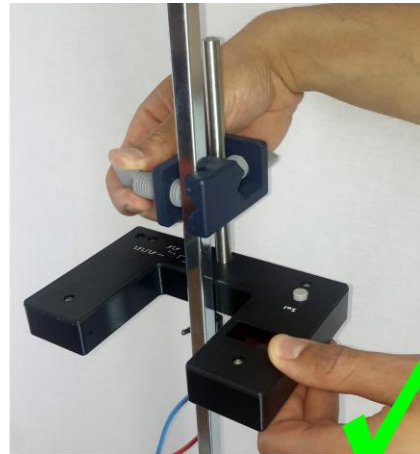
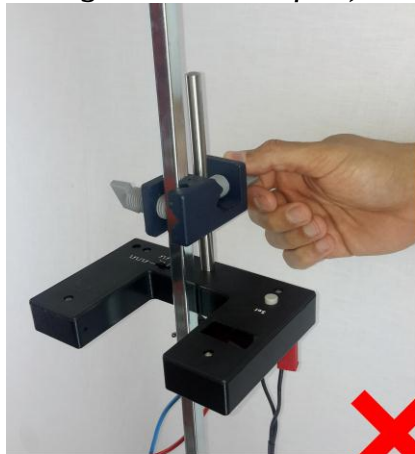


Figure 8.12 : Unscrewing the light barrier to adjust its vertical position.

- 6- Reset the light barrier using the grey reset button shown in Figure 8.13.
- 7- Unlock the release trigger (10) which in turn retracts the holding pin holding the wheel. The wheel is released and starts moving downwards. The counter of the light barrier should start simultaneously. Move quickly to the next step.



Figure 8.13: Light barrier reset button.

- 8- After the wheel has just passed the pin of the holding device on its way down, the release trigger (10) should be pressed again and locked before the light barrier is interrupted by the wheel.



The digital multimeter (11) buzzes until the release trigger is well pressed and locked. If it buzzes even after pressing on the trigger, check the wire connections or ask your TA to check your setup.

- 9- The counter stops as soon as the axis of the wheel cuts the light path of the light barrier. Record the distance S and the corresponding time t to cover this distance.



If you do not press the release trigger before the wheel passes through the light barrier, the light barrier will not measure the time. In this case, you should repeat your trial.



It is essential to watch the first down and up movements of the wheel, since incorrect winding (different densities, crossed over) will cause the wheel to have unstable motion. If so, repeat your measurement.

- 10- Now, to find the speed of the wheel after it travels the downward distance S adjusted in step 5, set the light barrier switch to the second mode ("▲▼") as shown in Figure 8.14.
- 11- Disconnect the "Trigger In" signal from the light barrier (3) by disconnecting the blue and red cords (12), (13) attached to the holding device (9) as shown in Figure 8.15.

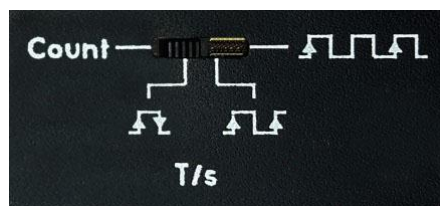


Figure 8.14: Set the light barrier switch to the second mode.

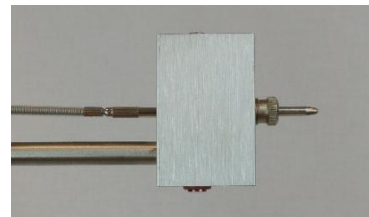


Figure 8.15 : The holding device after disconnecting the "trigger in" cables.

- 12- Repeat the steps from 3 to 6.
- 13- Unlock the release trigger (10) which in turn retracts the holding pin holding the wheel. The wheel is released and starts moving downwards. When the light barrier mode is adjusted to the second mode ("▲▼"), the light barrier time counter will not start until the wheel reached the light barrier.



With the current adjusted mode of the light barrier, the time measured will be the time Δt during which the orange cylinder at the tip of the wheel axis, shown in Figure 8.16, cuts the light beam of the light barrier on its way down.

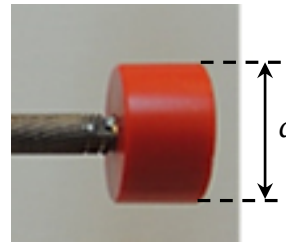


Figure 8.16: The orange cylinder at the tip of the wheel axis.

- 14- After the wheel passes through the light barrier and the time Δt is measured. Record the time Δt .
- 15- For each measurement, calculate the speed of the wheel v according to

$$v = \frac{d}{\Delta t}$$

where $d = 20 \text{ mm}$ is the diameter of the orange cylinder at the tip of the wheel axis shown in Figure 8.16 above

- 16- Repeat the steps from 1 to 15 with $S = 15 \text{ cm}$ several times for error analysis.



On repeating step 1, don't forget to reconnect the "Trigger In" signal to the light barrier (3) by connecting the blue and red cords (12), (13) to the holding device (9) as shown in Figure 8.5 (a) and (b) above.

- 17- Repeat steps 1 to 15 several times for error analysis for each of the following value of the distance S : 25 cm, 35 cm and 45 cm.
- 18- Plot the relation between the distance S , travelled by the center of gravity of Maxwell's wheel as a function of squared the time taken by the wheel to travel that distance (S vs. t^2).
- 19- From the slope of the (S vs. t^2) graph, calculate the moment of inertia of the wheel using relation (8.5).
- 20- Plot the relation between Maxwell's wheel speed after travelling a certain downward distance as a function of time taken by the wheel to travel that distance (v vs. t).
- 21- From the slope of the (v vs. t) graph, calculate the moment of inertia of the wheel using relation (8.6).



The mass of the wheel and its axis is $m = 0.504 \text{ kg}$ and the radius r of the axis around which the string is wound is $r = 2.5 \text{ mm}$.

8.5.2 Task (2): Verification of law of conservation of mechanical energy.

- 1- Use the data you got in Task (1) to calculate the change in potential energy and total kinetic energy (sum of rotational and translational energy) of the wheel.
- 2- Verify that the change in potential energy must be equal to the sum kinetic energies (translational and rotational) at every time point measurement.
- 3- Plot the change in potential, translational and rotational energies as a function of time.